

# A Packing of 62 Planes in $\mathbb{R}^{13}$

Limin Ge\*

October 8, 2026

## Abstract

We give 62 2-dimensional subspaces of  $\mathbb{R}^{13}$  whose smallest squared chordal distance is at least 1.700834. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 1.698904, obtained by deleting 1 of the 63 planes of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 62 subspaces of dimension 2 in  $\mathbb{R}^{13}$  whose pairwise squared chordal distances  $D(a, b) = 2 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 1.700834 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 13$ ,  $n = 2$ . The cell  $N = 62$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 1.698904, obtained by deleting 1 of the 63 planes of the packing found by this search. The Rankin simplex bound for this cell is  $2(13 - 2)/13 \cdot 62/61 = 1.720050$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n2N62.txt`

in <https://github.com/Ge-limin/math719>.

---

\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 114 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:

340166836494119445641903613957080328679522642279116090033664099188463164  
266769640068892136586230

denominator:

200000000000186569724511770066780675806266093282455628575071977680844100  
015551173608019965650459

## B Coordinates

62 blocks of 13 rows with 2 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m13n2N62.txt`.

|                             |                             |                             |                             |
|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| 422542411442 -80965514535   | 637606008754 264736267941   | -427429677496 227648710487  | 94052498577 -414571526393   |
| 116790785615 200639960814   | -19595844400 579220972334   | 97396107548 -22810737984    | -424494495104 252294250157  |
| 173634997287 200681231988   | -32216075928 330238446923   | 156125300751 -13815135743   | -236850689184 -40468959832  |
| -116749622451 -81882303033  | 216083505499 236625288043   | -13085759048 -245542583595  | -684035242984 -249246871600 |
| -149219306053 -215022659138 | -214344304527 -189967914536 | 247687421717 385701400950   | -190263671986 -277246789483 |
| -582699836763 309982619007  | -275958082750 139747502352  | -238283214924 471430119888  | -170702233002 -305605700795 |
| 148824531852 446755065134   | 341536088938 -406980449075  | 566287561171 167765257967   | 60864708081 37190380790     |
| -166437072767 90570920880   | 4021879161 54896224637      | -351094751984 -12108246374  | -130864277886 -212023169825 |
| 60339548143 -79700718232    | 249590347325 -5102264417    | 8433306417 484533066755     | 170980427628 337408994786   |
| -287160585521 381648283444  | -393695210036 -22269778160  | -469529212168 285122298009  | 267935675673 172004961520   |
| 2497999899 365979743983     | 83180841906 -437578787327   | -139292401664 535787453017  | -290338209917 290308092832  |
| -306334441049 110637264595  | -59058068117 153014133920   | 139940467595 201520976659   | 24743618919 -128683909448   |
| 415251251708 507842746692   | -193856089802 510117812411  | -92685572727 -228308587620  | -395142702789 -66037921051  |
| -526624172888 341084068591  | 366256935331 -30914371783   | 40507901036 161057963400    | 389900634365 493071644942   |
| -157185256466 -114110785294 | 340044562941 -241705197246  | -50782012699 -71823415229   | -23412538130 285362979026   |
| 51786198775 370234937799    | 105529399409 -129865206469  | -240082916902 -171662364633 | 175305921950 21142049262    |
| -324772905030 85436923636   | 373078431771 -149715492592  | -253644545225 -323708955808 | 460860696116 -348920352145  |
| 453782267325 250059414160   | -369865397045 -88522402322  | 147470964782 426995620691   | -74879938739 -330803617310  |
| 1322765801456 207295479555  | 418272391757 719806614420   | -81739755835 198276910493   | -470036567057 160180637457  |
| -14089083087 10356330781    | -111785885878 227558045485  | 244620692250 202013103174   | 136263167250 -352963665981  |
| -41079465832 -396670352984  | 138693158289 -170890858993  | 598924784162 -245774886390  | 331657533298 -1527387196    |
| -346743871705 -112496862835 | 26159378581 34612126776     | -398486734319 -225153852163 | -112936144027 162306127543  |
| 491107298296 20980348322    | 383064504313 -78119473461   | -347920378385 245871600043  | 182660780077 140174625206   |
| -44907331595 -332558997746  | 260627576348 58863502007    | -74489134550 426290253086   | -222452427626 60338420917   |
| -16949731937 -544176801014  | 203254084765 92953353813    | -172130867885 -252036936429 | 190068919 187038421530      |
| 25450808542 -208545667637   | -213170885103 -394753431488 | -68497362363 -32794568185   | -388151118987 -349044397891 |
| 108993247656 447198402483   | -392297382578 -118270172673 | 649447337667 233136111171   | -187986873798 -98374383948  |
| -591403953237 11230526977   | 464479227004 -56055011693   | -361344800025 -160728973262 | 488689688252 -113638426453  |
| 314722624386 1277165117626  | 110866632463 -465914829187  | 143691657442 -714390294303  | -363014374561 305463234769  |
| -80548914783 -157792464715  | 287157810577 -98590557932   | 297700246680 97164844634    | 126818794054 304511207984   |
| -33495166569 146881144720   | 149452590401 138214243438   | 14289157344 -43297415282    | 65353513818 -434041251165   |
| 368513630332 -99977681285   | 134514389607 -455280234092  | -11164014666 69683120918    | 420254456327 -255720895645  |
| -172541402947 -132230500956 | 580337164344 -66539385569   | -408739943474 203760174894  | -202072583318 -575446188173 |
| -258347905709 201097251745  | -86304008495 -24317136117   | -55840174316 -158741326466  | 230471057697 -7569165658    |
| -7188441789 -93846457437    | -127320814247 -219175768629 | -83318274765 -140703608601  | -271085813958 333065406946  |
| -265870797555 -276469606437 | 185904360057 -272813400165  | 178203001728 215382683185   | 305324579965 -6793835547    |
| 283134474786 -694454878518  | 115827107737 487581201078   | 48610382981 -76935639155    | -11791025810 92376819225    |
| -291246309821 -68224512471  | -79325773922 55858771708    | -146760655014 101112256804  | -209052343228 297620772513  |
| 252302361392 311702830097   | 40244587387 306514082641    | -27900303277 94069849964    | 67362430440 253085116540    |
| 567332099182 258512273071   | -205283522855 -322438927595 | -84264330248 -567143305759  | -108555812239 -324989433585 |
| -554896937441 270013713963  | -285213814104 533453325965  | -250154305987 366845102675  | 52312514911 167983084750    |
| -109525029052 -89956761767  | 141165689107 -261760262146  | 263246276888 148134151370   | 340016974267 543277056647   |
| 483841992422 264460344858   | 340914276685 -131038312824  | 106211271544 532647299996   | -471548131916 -108894538362 |
| 178015181515 -153819760653  | 633333612985 -11762609522   | 461649519363 65889547726    | -411720589391 26820391232   |
| 184208957819 -381712804607  | -14893715735 454741971287   | -62143828318 286545981097   | 453190886375 -212104348728  |
| 9341518499 -470673231069    | 253351445409 190322934482   | -244941142230 9651330574    | -66384456446 491170357068   |
| -75767489849 -321901684439  | 275233104608 387061660942   | 416808661357 -276355705251  | -356715810584 -479254453893 |
| -79463706515 399017090467   | -148285764161 156449039968  | -585319362348 -61798014979  | 558797873072 -116773505533  |
| 51824650671 56008639169     | -306026137635 -594844469125 | 375961941536 15612727470    | -300475781780 305452847326  |
| 188995575181 -261211601374  | 274836469197 87847909423    | -95340797764 -430450508911  | 223091223522 -266268578918  |
| 71736600232 204046689307    | 304781826722 115050416331   | 97525532654 -193226224652   | -246042328803 153549418897  |
| -52998194671 -124479332228  | -192627483572 -1490232538   | 415551486827 55784027999    | 49792655106 28252965156     |
| -162808853733 116527814540  | 346091925764 -181601908312  | -625138805480 -52154900568  | -107606200196 -718706944735 |
| -491912873627 517021636461  | -194532562946 64553227885   | -186643512271 -454361697091 | -110843330092 -34305636317  |
| -356714903797 109687832366  | 199620073378 32338315481    | 169492615368 74752546182    | -159192537049 -9528756748   |
| 17272912387 274460270147    | -119340917480 340361525829  | 167893387461 -371670367393  | 24037614465 182586309733    |
| -110123595079 -209459851928 | 438848851593 -318676235947  | -238894025299 396813421971  | -982717119890 125130897368  |
| 21999847681 299357688001    | -146892769468 493107433735  | 58538916789 24908193314     | -475662594279 -28320648683  |
| 310563615623 -126485957048  | 615523474178 -163164190531  | 20492618262 500004020477    | -268062557488 22319437025   |
| -140048212234 -81780970097  | -340680379214 -45754700140  | 348664161468 -107937840513  | 122721522720 48532203399    |
| -66951623970 112266066510   | 335879703256 663638905538   | 48315784901 37322540000     | 154005613399 -214302470235  |
| 653541946072 334929923797   | 33035953068 32474732078     | -438969087515 381869178361  | 290738829502 407434941289   |
| -196013937722 -291161815284 | -161082401415 -133515310992 | 107079936657 127847598854   | -217542803390 320508739180  |
| 18169330142 104698757462    | -72932048986 159042838265   | -28510165891 -154163827925  | 135312719107 284853229073   |
| -70324118595 -505030692174  | 387069873144 -202556546440  | -658314976 391134833180     | 297350427286 -38862225599   |
| -253677068690 108628600511  | 94076299154 -13908255673    | -41853605075 300200250450   | 12410990920 495427787129    |
| 134249157115 -36160979149   | 239020511715 431945412184   | 53420767439 316275762930    | 137343122204 414584127670   |

506926088183 153950774691 -539624257452 -153987882210 42532005402 596303529798 -29667481372 61314057889  
-278669005979 225143347967 -163764061268 200966089434 364484866522 35708789900 420307757290 -121395935174  
-406368437223 222390071785 -19321061244 -231256865631 30218221687 -369167782400 -389542679513 225165260204  
-295807317750 56915718946 90187520927 -228997817623 -188746171091 51301469878 131144789962 -377369598571  
-317972957269 236479504588 -356920388931 105951040390 192474055234 -482888501207 -486819121426 155029687648  
238766148777 -161689406279 104158253805 -34558596815 266456411506 200509499302 78981347583 193789555182  
45845659757 283909899640 365608033962 488029956848 -349708462194 329925504201 155148424714 436224777533  
601966519091 -357623361603 178338364657 187398084259 134621549280 163671302861 -64669011167 -41673830819  
29021556585 -30715264440 151962007749 -50070960190 354021905228 114264549622 -113298112255 460869702025  
180947717198 285463638693 -397956290462 219982500745 -389546593443 -218445257835  
-448532572240 -284845422841 -249883613837 188976493667 -460234453549 -375400651742  
-37793071620 -133947206470 -240033070649 248802506997 -460778034191 -202503432028  
-113328812378 -33718504836 296287454905 357798423507 49625488662 613111795340  
-91775003972 29006097192 483388567747 38830358951 403392542798 23942461891  
393771592666 -37143935624 44024528292 -269936689061 222264186866 -248549473760  
-26543056843 -323661453457 -295104411876 -326257376081 373966023430 -15622724094  
324012712892 -221110705113 -431538772869 51431415920 -326062907534 574650472179  
245601686905 577444054919 48475963601 114984764918 358763367507 200709815306  
-20054589514 93211773753 588265516015 107287502563 -354667582615 -120785568080  
-481231896398 482701604672 176415628240 -594928983753 -142004780444 69383594832  
-61136230064 193368272237 74973050794 176888412777 126972209797 151848582103  
-451861740207 -111447020661 100007142636 -532815924429 110287929623 131820201544  
-25361028174 39551867908 37877102539 -344932978907 3350829257 -172879746675  
-64118143249 -165473863018 -464931737093 -359621732524 -154803721383 -189994397961  
283648933456 276928816544 -463883097931 468936641580 -8559268573 -7200293183046  
614856483976 201498928727 279462254108 -80716763112 211922805594 159505764948  
190226592929 -559265606418 222172705968 -190630955389 43245923408 -262092935782  
139643164737 435987643211 -109415343708 -28519531432 -82712726922 4247930575  
-116190784573 -109354632598 -301343075423 -275001354434 -2405681497 159173483493  
123425526797 47694750140 -51653072749 -40525162520 81889129632 164957016295  
25843229478 212522468782 -58582788638 130140986014 -30778940608 305760826892  
-310999674833 511761996401 -2490997872 19106055124 -544225919339 85744417934  
259598031496 263675392996 115085474784 -343575751420 -406002688316 363148783345  
186007342748 17807071629 -552510838233 -181078345372 165865597388 -223137103988  
264295976450 -251876591695 112572828987 -282685993608 -403144081192 -110765048025  
565336692311 178722621089 15912015680 -170902326440 400353455609 167098831147  
146799341827 -259032743232 57049746961 282278552998 -346494301208 -101840889591  
147845275648 406791882156 418935278895 332130437643 -98440081667 -6338469015  
321698200843 -334779981864 -320361140190 88676840685 61022703858 70619999614  
16191274473 116269283257 481269664364 86941114890 -372008675229 395495880522  
467617162280 325143735539 209467185386 326083868009 -1397577679 -268823951774  
-169632983671 -65062944865 -51782284862 326216859094 -279534781817 -516691183305  
12293028823 277626856763 448115948013 -190089669445 21152093888 -133941557341  
-46096829726 -166349555286 -91064581419 171267359606 -21276226685 -42048256281  
564732604264 31008142995 228255580768 -366926929045 147302492180 280025896015  
-334179181385 194424833751 140005263387 482194117512 162110810391 119705906719  
-109328000264 -29882918936 -461765528599 -184141948889 -67886222343 107570998022  
-367217117659 324303090107 268497190390 73871798179 -70563853728 320486723749  
-341803102843 7462303734 92979902856 36206490746 326479460842 30478745218  
275193821053 171472214725 -61467557099 655387785379 -15147326175 -26065072405  
140047180552 -187584113545 805425323202 -67738684715 -25343573959 -14054772292  
33145003404 -11512330058 -351330869469 289009414763 -119719534008 -227765121339  
-96579201123 -84912124556 464448845293 268539602608 81844338677 294257658717  
245741749814 245098406155 280499120167 -212072386739 759593425117 -123063701044  
-254423644841 388146469297 65564656251 277546617299 145936294173 -28393380111  
231550353714 756600020740 95590851530 -218695160891 -167672276049 -260819538278  
-145821897961 -2947131895 -166255921847 415102388987 -119859318393 41717736584  
-58769300745 -295778012880 144617983427 -228336353806 61474281449 304341681290  
-32751654010 132614657146 65842657259 -23286127464 258567204195 243137900837  
48976482942 -421461864388 -54383680900 -32703380027 163528686394 -379650851880  
1324443020505 -31473035372 -361415796413 -114165745332 -147888000430 351247924984  
-10116285814 55307754229 -130404561753 124288360953 334109242879 316917279983  
-388406127621 48997866047 99085313882 110125703449 -207607011178 -61844315834  
388278774704 251215589938 -6608828535 291553029109 -221240912200 -135956964338  
-124326035865 744870507267 -350620099467 -677682422284 247394426900 -380846126535  
-439236696294 -125027070431 122305752315 336579823840 -107872883707 -676900248033  
-465853789442 -47738683078 -365176954037 230292449362 404163291781 3995375502  
399158868385 -15740176183 119963359216 -87648820754 -133503841486 254672805041  
270645214006 -38387001038 251024639042 2720589990 -58568306470 -272911641252  
125814456824 -270693398037 351151158144 117748110725 -197388552842 -261005970181  
-515027830286 213900140547 -575081144543 388410438483 319427758604 16074867454  
-54813854609 -75133464972 -355888646536 -3789314201 359891732051 268265912756  
115934269749 782906033800 169789652408 186432104606 -68330833545 85165117264  
155016339765 -332821689592 122353687479 -233793105643 -407555460756 22522153367  
46018398171 -43378622088 -200408290383 84430696005 -413248195491 299411538044  
96511459571 -216691294343 626615744572 153432247661 161945671557 -34150986104  
59102647071 73709582283 -89178312358 497038394726 -179895332407 124921518699  
81248250151 311705745344 -259233673148 312685204107 -481996818539 -71989465771  
-594272459032 -17621061464 27103733570 -163199012211 319904588506 -151682153469  
-295665708486 -160372359820 245945335783 -171416585994 -65816800912 282638985729  
-6943180480 -207869424594 168269587952 195136770099  
117989493374 44468172968 -149175726207 815085972528  
-48407760361 47352399134 -76011575699 110507727564  
-178379600957 -327800678052 227242078435 5239551686  
-339260494269 -167295789411 484285966903 -119416981844  
-317317269718 347439418061 20285675537 178814466232  
-47376857845 327668307421 -217912671738 -184523345747  
-163559017925 -91646392109 -313216332478 -170466769235  
-79322985546 -229406660354 404523335667 228679105410  
704591734575 124408186754 12012076817 -81464264072  
-50756936170 120541697761 352814054259 123720864611  
112926559927 -421385511836 193987942585 -40489277915  
-88594244877 141355983952 44463682000 -91515436481  
507061688951 -233224733251 -54016156334 42088892624  
-196696157707 -489988182155 -175824527991 -302477692176  
124992697469 23111584093 292400754435 -308719779149  
171008924823 190407546029 -554744452506 10894620105  
35687947033 460623066479 -74788063521 -130053684030  
74119635439 81102092207 349488305052 -22924580861  
-461672051395 468851390414 -3133243712 -171617596453  
-330854265776 -150393697994 71545178785 779203470292  
-207249921081 286254061105 155720960746 270693007984

|                              |                             |                             |                             |
|------------------------------|-----------------------------|-----------------------------|-----------------------------|
| 325510390301 -226866209802   | -347280076996 265595389500  | 273193641259 452357374386   | -244545148345 286904381602  |
| -86472660533 143957535109    | 91194269020 -295384354728   | -139830560231 -581335804068 | -228001130940 364980399286  |
| -180921526791 -424774185886  | 33956154969 311841159667    | 148095011260 201709834645   | -75002349291 -33511462390   |
| 455114044797 113737624352    | -194704310548 299610177827  | 388310885922 -168928672695  | 8787290688 -143258640232    |
| 260418382709 249674259664    | -256670441090 -253082927292 | 259992236375 -126214332891  | 412855531661 228824939258   |
| -489688172570 -196169291036  | -220384087668 -519577739289 | -195839260743 -303685273830 | 129666604500 79591751629    |
| 238143358058 -206413204862   | -78097664204 141625772628   | -438883795424 235481111241  | 224684807770 -203126756906  |
| -152938848323 93306453842    | 333430843296 99129127133    | 447260208131 -227992341817  | 25232342616 -71106917237    |
| -195461204853 81580105425    | 89808436075 233108780150    | 44673334726 -354855751460   | 226440883954 76207820069    |
| 402779482160 -454732975711   | 159309675686 543315740635   | -97816962419 -122994889729  | 192850201663 -240472644717  |
| 38612922458 436899496104     | -90521554762 -33359383366   | 394287584650 80207948580    | -620626311442 -386835483872 |
| -145036337976 196916294817   | 30034664872 186064380083    | 154250201759 72450311939    | 277759833143 118404593712   |
| -189806191866 -3731111073184 | 139301646345 562702482166   | -480469384520 -370585302780 | 509286751203 -130424193316  |
| -194418366249 -392895032092  | 83582163510 -269325419894   | 396218356 -356199559624     | -12809258098 504481366734   |
| -117081490282 -13236916577   | 714019563808 -391666670810  | -209217221309 21381269739   | -17047879952 268258740752   |
| -132289488794 -471095886482  | 370828630874 127813514730   | -432699531818 -158111555267 | 92902899561 -310283666904   |
| 455923671210 -76445723270    | 161354529591 -83093078202   | -67744899821 -234891439392  | -275216703960 495486005372  |
| -121445882558 -537856299839  | 374671585777 137960651175   | 256270129713 -117779784507  | 192534503434 186535275191   |
| -35663200507 286534916434    | 6359838124 2028683526       | -49287081809 -197839759994  | 69147859184 -88985801495    |
| -575538783034 338854160130   | 442974428976 -66075951426   | -513300152623 414998910405  | 11399102245 198322195327    |
| -230191573498 -18885160835   | -218722554491 -31390024310  | -89494957223 -25874654077   | -276413371618 -425271014397 |
| 124420967549 -64621621166    | -407359245103 -162200820473 | 283412601530 -167824831264  | -273308100898 -205324609723 |
| -42491859088 122592031927    | -70093317691 -119195314057  | -25538974395 -605113238173  | 116612731668 164248588754   |
| -78132822238 -107329872914   | -389478624516 -185959709610 | -56933357731 -186239564111  | 494220958295 -340090822954  |
| 65297041501 315727499414     | 87479395444 -44037213671    | 331289258145 -15471726059   | 260634320764 -210174606389  |
| 543339604983 26537340887     | 225646343812 -197678156424  | 426197883104 -584219337419  | 242926345653 -436140768087  |
| 599300881706 272060174688    | -99112922225 363960320816   | 183383130358 -68778612313   | 261769595663 104977177176   |
| -146178882202 349663279181   | -336842715661 131675833201  | -369860919901 -214209454319 | 163919622182 566322850040   |
| -485232044082 40013779503    | 108068694720 570793811904   | 222128437468 265802052700   | -271115920855 -105906440149 |
| -175646199914 244113926100   | 408630735271 -249660895825  | 105763157434 -10285566769   | 503866058545 18573817904    |
| 142491504706 -204794961390   | -216917732616 199805707720  | -206292390588 -442910899019 | 190821407056 -96962685727   |
| 47163854505 -43693426685     | 160683993791 548334718052   | 94437907583 170849569829    |                             |
| -219564815799 140925720209   | -208963691832 140990637596  | -497983765471 -149745413572 |                             |